

A Composition-Driven Machine Learning Framework for Crystal Structure Generation via Prediction of Formation Energy, Stability, Volume, Space Group, and Wyckoff Configurations

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Abstract—The challenge of finding stable crystalline materials has been one of the major challenges in computational materials science and is typically solved with expensive first-principle calculations and crystal structure prediction methods. Recently, ML-based approaches have shown potential for speeding up the materials discovery process; however, the majority of methods have been either reliant upon known crystal structures for input, or focused only on isolated prediction tasks. In this study, we propose a composition-based ML framework to provide end-to-end predictions of key structural and stability properties of crystalline materials. Our framework utilizes multiple models to predict formation energy, thermodynamic stability, volume, space group symmetry and Wyckoff position configurations directly from chemical composition. We have trained and evaluated regression and classification models for all of these tasks using a curated dataset from the Materials Project, and demonstrated competitive performance across all phases of the analysis, including good accuracy for space group prediction and reasonable recall for certain Wyckoff patterns, despite the large and unbalanced label space. Furthermore, we describe how to combine the results from the various predictions to propose likely crystal structure candidates and show how to connect composition to structure without prior knowledge of these structures. Overall, this modular and interpretable framework provides a scalable approach for rapid screening and design of novel materials, bridging the gap between composition-based prediction and crystal structure generation.

Index Terms—Machine learning, Composition-based prediction, Formation energy prediction, Thermodynamic stability, Energy above hull, Space group prediction, Wyckoff positions, Multi-label classification, Materials project

I. INTRODUCTION

The development of innovative crystalline materials is key for ongoing growth in technology such as energy storage, electronics, catalysis and structural engineering. One of the foremost challenges in materials science is accurately predicting both the thermodynamic stability and crystal structure of a material based on its chemical makeup alone. Historically, this has been

accomplished by performing ‘first-principles’ calculations using methods like density functional theory (DFT) in conjunction with crystal structure prediction (CSP) techniques such as evolutionary algorithms and random structure searches. Although effective, these methods can be costly in terms of computational power, thus limiting large-scale material design space searches [1], [2].

The rapid advancement of machine learning (ML) is now allowing researchers to predict many material properties much more quickly, thus providing an alternative method to traditional computational methods that can then be scaled up accordingly. Various types of ML techniques, including graph neural networks and composition-based method, have shown success at predicting properties such as formation energies, band gaps, and thermodynamic stabilities. However, many current techniques utilize known crystal structures as inputs or attempt to predict single properties independently. Because of this, these techniques cannot be used during the early stages of discovering new materials when only the chemical composition is known [3], [4], [5], [6].

Crystal structures can be described on a compact basis in terms of multiple thermodynamic or structural descriptors, including formation energy, energy above hull, lattice volume, space group symmetry, and Wyckoff positions. These descriptors characterize the stability, symmetry, and atomic arrangement of crystalline materials [7], [8], [9], [10]. As important as these properties are, very few studies have attempted to perform composition-only predictions of these properties together in a cohesive manner.

Simultaneous prediction of these properties is challenging due to the need to perform both regression and classification tasks; the highly unbalanced label spaces for symmetry and Wyckoff positions; and the existence of highly non-linear relationships between composition and material functionalities [13], [14],

[15], [16], [17]. Therefore, a unified framework is required to integrate thermodynamic prediction with structural descriptor prediction in order to provide a more complete approach to predicting crystal structures from composition only.

The machine learning framework we propose is driven by the compositions of materials to predict the various properties of materials, including formation energy, energy above the hull, thermodynamic stability, volume per atom, space group symmetry, and Wyckoff position configurations. The integration of several machine learning models in this hierarchical framework allows for the sequential prediction of thermodynamic and structural descriptors in order to ensure that the predictions made by the framework are physically consistent with each other. Therefore, the proposed composition-driven framework for predicting crystal structures can be used to characterize the crystal structures of materials on a large scale without having to perform either DFT calculations or predict the crystal structures of the materials.

The primary contributions of this research are:

- 1) We develop a multi-task, composition-derived machine learning framework for simultaneous prediction of formation energy, energy above hull, and thermal stability.
- 2) We introduce a physics-informed volume prediction model with uncertainty estimation for predicting lattice volume from composition.
- 3) We develop machine learning models for predicting space group symmetry and Wyckoff configurations using composition-based descriptors and symmetry constraints.
- 4) We propose a hierarchical prediction pipeline will sequentially predict thermodynamic properties, volume, symmetry and Wyckoff positions in order to build physically consistent descriptors of crystal structures.
- 5) The prediction models will also provide uncertainty quantification and risk control to give a higher level of reliability for materials discovery efforts.
- 6) The proposed framework provides a scalable approach for connecting chemical composition to crystal structure descriptors without requiring crystal structure prediction or DFT calculations.

II. DATASET AND PREPROCESSING

A. Dataset Construction

In this research, we utilized data from the Materials Project database to create a comprehensive database of inorganic materials. The Materials Project is considered a well-established and widely used resource for data-driven research in materials science as it provides a number of density functional theory (DFT) calculated properties for both experimentally known materials and hypothetical materials.

For each material entry, the following properties were extracted:

- Chemical composition (formula and elemental fractions)
- Formation energy per atom
- Energy above hull (thermodynamic stability indicator)
- Volume per atom
- Space group number
- Wyckoff positions (letters and multiplicities)

Furthermore, structural metadata was also kept for further processing, such as the number of atoms in the unit cell as well as the objects that hold the data on composition.

B. Data Cleaning and Filtering

Multiple preprocessing steps were performed to provide consistency and reliability to the dataset:

- 1) **Removal of incomplete entries:** If any material had missing values for any of the required target properties (formation energy, energy above hull, volume, space group, or Wyckoff information), that material was removed from consideration.
- 2) **Standardization of compositions:** The conversion of chemical formulas into standardized composition objects is meant to provide a consistent way to represent chemical formulas for all materials that are in the database.
- 3) **Filtering of rare Wyckoff patterns:** Wyckoff configurations that occur infrequently have been removed to address class imbalance and to improve the generalization of the models. This resulted in the deletion of unique classes of Wyckoff topologies and the retention of only the most representative structural patterns.
- 4) **Consistency checks:**
 - Alignment between element lists and fractional compositions
 - Verification of element–Wyckoff mapping
 - Removal of duplicate or inconsistent entries

C. Target Variable Preparation

Different target variables were prepared depending on the prediction task:

Regression Targets:

- Formation energy per atom
- Energy above hull
- Volume per atom

Classification Targets:

- **Stability label:** Materials were classified as stable if their energy above hull was below a predefined threshold (typically close to 0 eV), and unstable otherwise.
- **Space group:** Treated as a multi-class classification problem using the space group number.
- **Wyckoff patterns:** Encoded Wyckoff Letters will represent the Symmetry Equivalent Atomic Sites

and be treated as either Categorical or Multi-class output depending on the Modeling Strategy employed.

D. Task-Specific Dataset Preparation

As the prediction tasks contained in this study consist of both regression and classification problems, as well as having different target output spaces; therefore separate datasets were created specifically for each task using data from the master dataset after it was cleaned.

- For formation energy, energy above hull, and volume prediction, datasets were prepared with continuous target variables suitable for regression models.
- For stability predictions, a classification dataset containing binary classification labels that were derived from the calculated stability level.
- For space group prediction, a multi-class dataset was constructed using space group indices as labels.
- For Wyckoff pattern prediction, a specialized dataset was created that included an encoding of the Wyckoff configurations, including handling multiple Wyckoff sites per material and reducing rare classes to manage label sparsity.

This preparation specific to the task ensures that each model receives training information that is specific to meet its' learning objectives.

E. Dataset Splitting

Each dataset has been divided into two sub-datasets; training and testing; in order to allow the assessing of model's performance. A standard split of 80:20 training to testing has been applied so as to ensure:

- No data leakage between training and testing sets
- Representative distribution of compositions and target variables
- For classification tasks, class distribution was preserved across splits where applicable.

F. Summary

The final processed datasets consist of composition-based inputs along with task specific target output variables, which enable:

- Regression tasks for thermodynamic and structural properties
- Classification tasks for stability, symmetry, and atomic configuration

This structured and modular method of data preparation is the basis of the multi-task machine learning framework that has been developed through this work.

III. FEATURE ENGINEERING

A. Composition-Based Representation

The models developed in the current study are using machine learning to formulate predictions about materials based only on the chemical composition of the materials without using atomic coordinates or

crystal structure information as input. The composition of each compound was transformed into a normalized fractional representation, by finding the fraction of each element that exists in the compound. This provides a representation of input to the machine learning model in such a way that it does not depend on the absolute number of atom types but only on the relative proportions of each element type.

Because composition is the only attribute available for most early-stage materials, this composition-based representation allows for the application of this framework to the discovery of materials in early stages of development when data about the structure of the material is still not available.

B. Elemental Property Descriptors

We have converted chemical compositions into numerical feature vectors that can be used in machine learning by creating a set of compositional descriptors according to elemental characteristics. The set of these descriptors was developed to encapsulate the statistical distribution of elemental characteristics in each chemical composition and provide a common length to represent a material regardless of its number of elements.

Elemental properties considered include: Atomic number, Atomic weight, Electronegativity, Atomic radius, Covalent radius, Valence electron count, Ionization energy, Electron affinity, etc. These elemental attributes provide physically meaningful information related to bonding behavior, atomic size, and electronic structure, which are important factors governing material stability and crystal structure.

C. Magpie Feature Set

Magpie descriptors were used to create feature vectors, which are typically used in materials informatics to predict properties based on their compositions. For each elemental (periodic table) property, Magpie computes a set of statistics from the elements that make up a sample (element, range of values). When the same calculations are performed for multiple types of elemental properties, this produces approximately 130-140 compositional descriptors for each sample [3], [4], [11]; thus, each sample will have a fixed-length feature vector.

Through the use of these descriptors, machine learning models can learn to correlate the relationship between composition and the resulting material property, allowing machine learning models to learn the relationships between composition and properties.

D. Feature Normalization

Feature scaling was applied before training the models, allowing every feature to contribute to the learning process by having equal contributions to the model's final output. The way that this was done was by standardizing all of the features of each of the materials in the dataset to have a mean of zero and

a variance of one, using the mean and variance that were calculated based on the training set of materials.

This improves the overall numerical stability and performance of the models, and helps with model performance on algorithms that are sensitive to the scale of their features.

E. Final Feature Vector

The representations of every material are now in the form of a single fixed-length feature vector that consists of approximately 130 - 140 Magpie descriptors that were derived from the materials' compositions. Each of these feature vectors will be used as inputs to each of the machine learning models being developed in this work for predicting formation energy, predicting energy above the hull, predicting stability class, predicting volume, predicting space group, and predicting Wyckoff pattern.

This unified feature representation enables consistent learning across multiple prediction tasks and forms the basis of the multi-task machine learning framework proposed in this study.

IV. METHODOLOGY

A. Overview of the Framework

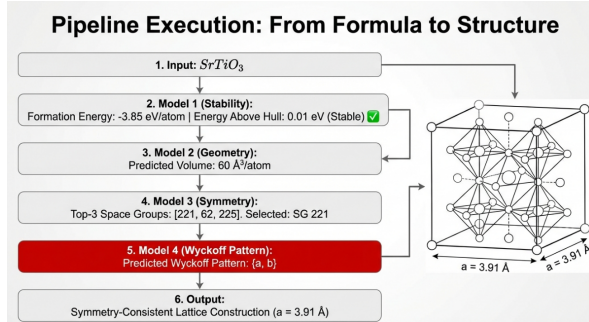


Fig. 1: Overall workflow of the proposed composition-driven crystal structure prediction framework.

In this work, we develop a composition-driven machine learning framework for predicting both thermodynamic and structural descriptors of inorganic crystalline materials. The framework predicts formation energy per atom, energy above hull, thermodynamic stability, volume per atom, space group symmetry, and Wyckoff position configurations directly from chemical composition as shown in Figure 1.

The overall mapping learned by the framework can be expressed as:

$$C \rightarrow \{E_f, E_{hull}, S, V_{atom}, SG, W\} \quad (1)$$

where C denotes the chemical composition, E_f is the formation energy per atom, E_{hull} is the energy above hull, S is the stability label, V_{atom} is the volume per atom, SG is the space group, and W represents the Wyckoff configuration.

Instead of predicting full atomic coordinates directly, this work focuses on predicting physically meaningful intermediate descriptors such as thermodynamic stability, lattice volume, symmetry, and Wyckoff configurations, which together define the crystal structure.

The framework consists of four separate machine learning models, each designed for a specific prediction task, as summarized in Table I.

TABLE I: Models and prediction tasks

Model	Predicted Properties	Learning Type
Model 1	Formation energy, Energy above hull, Stability	Multi-task learning
Model 2	Volume per atom	Regression
Model 3	Space group	Multi-class classification
Model 4	Wyckoff configuration	Multi-label classification

A hierarchical prediction strategy is used in this work, where thermodynamic properties, volume, symmetry, and Wyckoff configurations are predicted sequentially to ensure physically consistent structural descriptors.

B. Multi-Task Thermodynamic Prediction Model

1) *Composition Representation*: Each material composition is represented using its constituent elements and their normalized atomic fractions:

$$C = \{(e_1, f_1), (e_2, f_2), \dots, (e_n, f_n)\} \quad (2)$$

Each element is mapped to a learnable embedding vector, and a composition representation vector is obtained using fraction-weighted pooling:

$$\mathbf{z} = \sum_{i=1}^n f_i \cdot \mathbf{E}(e_i) \quad (3)$$

2) *Multi-Task Neural Network*: The composition vector is passed through shared dense layers to learn a common thermodynamic representation:

$$\mathbf{h} = f_{shared}(\mathbf{z}) \quad (4)$$

Separate output heads are used for formation energy, hull distance, and stability:

$$E_f = f_{FE}(\mathbf{h}), \quad E_{hull} = f_{hull}(\mathbf{h}), \quad S = \sigma(f_{stab}(\mathbf{h})) \quad (5)$$

The model is trained using a multi-task loss function:

$$L = L_{FE} + L_{hull} + L_{stab} \quad (6)$$

$$L_{FE} = \text{MSE}(E_f^{true}, E_f^{pred}) \quad (7)$$

$$L_{hull} = \text{MSE}(E_{hull}^{true}, E_{hull}^{pred}) \quad (8)$$

$$L_{stab} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (9)$$

3) *Model Variants Used:* There were several different types of model variants created and compared throughout the study. Each model was built under a multi-task learning framework where predictions for formation energy/atom, energy above the hull, and thermodynamic stability could be made simultaneously based on composition.

The TC-MTL model was created out of both element embeddings and transformer encoder layers to allow interactions among elements within a particular composition to be captured. In addition, a lightweight transformer encoder model (LTC-MTL) was created to reduce the computational complexity due to this model's use of fewer layers and smaller embedding dimensions than TC-MTL. In order to improve the robustness of the predictions, an ensemble of LTC-MTL models (E-TC-MTL) was trained independently, and their predictions averaged to provide one final prediction for all three target properties. Finally, an optimized lightweight transformer model (O-LTC-MTL) was created with the introduction of learning rate scheduling and early stopping to facilitate convergence.

In addition to the embedding-based models described above, a feature based model with approximately 130-140 total Magpie descriptors (FB-MTL) was also implemented. Finally, an uncertainty-aware ensemble model (UQ-MTL) was developed to provide an estimate of uncertainty around the predictions and allow for controlled risk prediction.

C. Volume per Atom Prediction Model

1) *Target Definition:* Volume per atom is defined as:

$$V_{atom} = \frac{V_{cell}}{N_{atoms}} \quad (10)$$

2) *Feature Representation:* The volume model uses:

- Composition embeddings
- Physics-informed descriptors:
 - Mean atomic radius
 - Std atomic radius
 - Mean electronegativity
 - Mean valence electrons
 - Mean atomic mass

These are computed as:

$$\bar{p} = \sum_i f_i p_i \quad (11)$$

3) *Model Used:* The volume model is a neural network with uncertainty-aware regression, consisting of:

- Element embedding layer which maps chemical elements to dense vector representations.
- Composition encoder processes the fractional composition of the material.
- Physics feature encoder incorporates the stoichiometric averages of elemental properties.

- Fusion network combines structural and compositional features for high-level representation.
- Mean and variance prediction heads outputs both the predicted value μ and the associated uncertainty σ .

The model is trained using Gaussian negative log-likelihood loss:

$$L = \frac{(y - \mu)^2}{2\sigma^2} + \frac{1}{2} \log \sigma^2 \quad (12)$$

D. Space Group Prediction Model

Space group prediction is formulated as a multi-class classification problem using Magpie composition descriptors (138 features).

TABLE II: Models Used for Space Group Prediction

Model	Description
Random Forest	Tree-based ensemble
XGBoost	Gradient boosting
MLP	Neural network
RF Ensemble	Ensemble of two RF models
RF + Crystal System	RF with symmetry filtering

The final probability is computed using ensemble averaging:

$$P(SG) = \frac{1}{N} \sum P_i(SG) \quad (13)$$

Crystal System Constraint: A separate model predicts the crystal system, and space groups inconsistent with the predicted crystal system are removed. This symmetry filtering approach significantly improves the accuracy of the prediction by ensuring physical consistency within the classification output.

E. Wyckoff Position Prediction Model

Wyckoff position prediction is formulated as a multi-label classification problem. Magpie composition features and space group information are used as input.

TABLE III: Models Used for Wyckoff Position Prediction

Model	Description
LightGBM	Gradient boosting framework
XGBoost	Gradient boosting framework
Multi-label binary classifiers	One independent classifier per Wyckoff letter

The model predicts the probability of a Wyckoff position w_i given the composition C and the space group SG :

$$P(w_i | C, SG) \quad (14)$$

The model is optimized using the binary cross-entropy loss function:

$$L = - \sum [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (15)$$

To ensure physical validity, the predicted Wyckoff positions are filtered using the following criteria:

- **Space group constraints:** Ensuring the predicted positions are allowed within the specific symmetry of the space group.
- **Wyckoff multiplicity rules:** Validating that the site multiplicities are consistent with the stoichiometry.
- **Atom count constraints:** Matching the total number of atoms in the unit cell.

F. Separation of Machine Learning and Physics-Based Structure Construction

The proposed system of work will use machine learning to predict thermodynamic properties and structural properties, while also using crystallographic symmetry operators and lattice constraints to create the final crystal structure. Instead of predicting atomic coordinates directly, the model predicts space group symmetry, Wyckoff positions, and lattice volume, and atomic coordinates are generated analytically from symmetry operations. By separating the machine learning prediction from the physics-based crystal structure creation, it is ensured that the structures generated through this methodology will satisfy crystallographic symmetries and stoichiometric properties.

G. Physically Conditioned Wyckoff Blueprint Representation

Based on anticipated thermodynamic and symmetry descriptions, this work presents a physically validated structural representation of crystals. This representation consists of predicted space group symmetry, Wyckoff site configuration, elemental mapping to Wyckoff sites, and lattice volume prediction. Collectively, these descriptors characterize the crystal structure in a way that is physically valid. Lattice parameters are computed from predicted lattice volume and the crystal symmetry, and atomic coordinates are computed based on Wyckoff positions and symmetry operations. With this representation, the crystal structure can be constructed using the known composition, without predicting atomic coordinates directly from composition alone.

H. Hierarchical Prediction Pipeline

The predictions are performed sequentially:

$$\begin{aligned} \text{Composition} &\rightarrow \text{Thermodynamic Properties} \rightarrow \text{Volume} \\ &\rightarrow \text{Space Group} \rightarrow \text{Wyckoff Positions} \end{aligned} \quad (16)$$

This hierarchical framework ensures physically consistent structural descriptors and reduces the search space for crystal structure generation. Additionally, uncertainty-aware learning and crystal symmetry constraints are incorporated to improve prediction reliability and physical validity.

V. RESULTS AND DISCUSSION

A. Thermodynamic Property Prediction

This section evaluates the performance of different machine learning models developed for predicting formation energy per atom, energy above hull, and thermodynamic stability using composition-only inputs. Multiple architectures and training strategies were explored, including transformer-based models, feature-based models using Magpie descriptors, ensemble models, and uncertainty-aware models. The dataset consisted of 150,200 materials, which were split into training (80%), validation (10%), and test (10%) sets.

TABLE IV: Comparison of Thermodynamic Prediction Models

Method	Formation En- ergy MAE (eV/atom)	Hull Dis- tance MAE (eV)	Stability F1 Score
TC-MTL	0.139	0.076	0.898
LTC-MTL	0.120	0.071	0.896
Ensemble Transformer	0.129	0.082	0.888
Optimized LTC-MTL	0.104	0.066	0.909
Feature-Based MTL (Magpie)	0.103	0.059	0.819
Uncertainty Ensemble	0.0975	0.0749	0.9306

The performance of different thermodynamic prediction models is summarized in Table IV. As shown in Table IV, the uncertainty-aware ensemble model achieved the lowest formation energy error and the highest stability F1 score, while the feature-based multi-task model achieved the lowest hull distance error. These results indicate that multi-task learning combined with ensemble methods improves thermodynamic prediction performance and reliability.

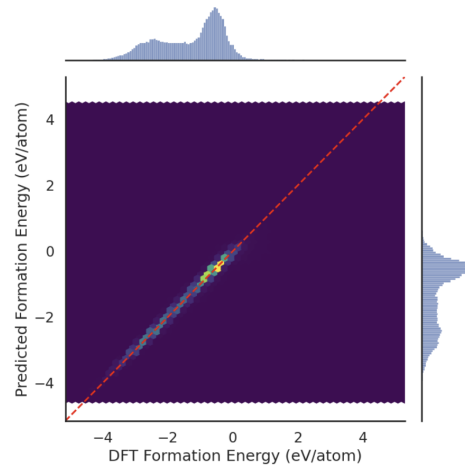


Fig. 2: Parity plot comparing predicted and DFT-calculated formation energies.

The parity plot comparing predicted and DFT-calculated formation energies is shown in Figure 2.

Most data points lie close to the diagonal line, indicating strong agreement between predicted and true values.

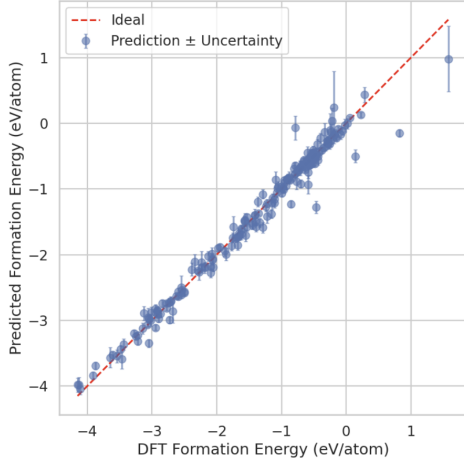


Fig. 3: Ensemble prediction with uncertainty bounds.

The ensemble prediction results with uncertainty bounds are shown in Figure 3. Predictions with larger errors generally exhibit higher uncertainty, demonstrating that the model can estimate prediction confidence.

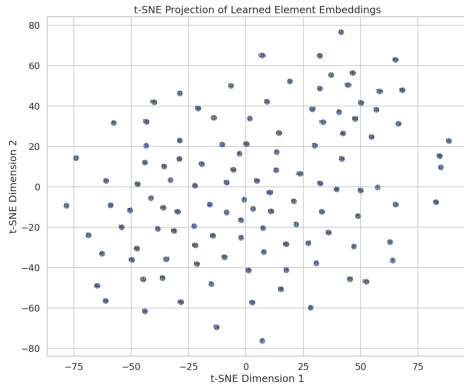


Fig. 4: t-SNE projection of learned element embeddings.

The t-SNE projection of learned element embeddings is shown in Figure 4. Elements with similar chemical properties cluster together, indicating that the model learns chemically meaningful representations from composition.

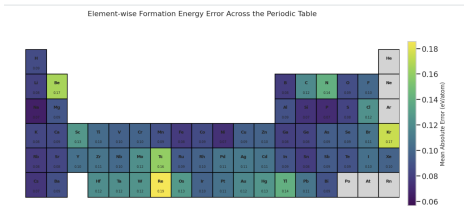


Fig. 5: Element-wise prediction error heatmap.

The element-wise prediction error distribution is shown in Figure 5. Higher errors are observed for ele-

ments with limited training data and complex bonding environments.

B. Uncertainty Quantification and Risk-Controlled Prediction

To provide an estimate of uncertainty instead of solely relying on point predictions for estimating uncertainty, an ensemble of separate, independently trained models has been used. The average prediction of the predictions made by the different members of the ensemble was taken as the final prediction and the average standard deviation of the predictions made by the independent members as a measure of epistemic uncertainty.

The uncertainty estimates were evaluated using the measurement of how closely the uncertainty of the predictions correlates with absolute prediction errors. If the correlation is positive, then the model will be able to identify these uncertain predictions.

TABLE V: Uncertainty Model Performance

Metric	Value
Formation Energy MAE	0.0975 eV/atom
Hull Distance MAE	0.0749 eV
Stability F1 Score	0.9306
ROC-AUC	0.9557
Uncertainty-Error Correlation	0.6396
FE MAE @ 10% Coverage	0.0446 eV/atom

As shown in Table V the uncertainty-aware ensemble model achieved a formation energy MAE of 0.0975 eV/atom and stability F1 score of 0.9306. The uncertainty-error correlation of 0.6396 indicates that prediction uncertainty is strongly correlated with prediction error. When only the lowest uncertainty predictions (10% coverage) were considered, the formation energy error decreased significantly to 0.0446 eV/atom, demonstrating the usefulness of uncertainty estimation for risk-controlled materials screening.

C. Volume per Atom Prediction

The volume per atom prediction model was trained on 151,816 materials using composition embeddings and physics-informed features such as atomic radius, electronegativity, valence electrons, and atomic mass.

TABLE VI: Volume Prediction Performance

Metric	Value
Test MAE	0.640 Å ³ /atom
Uncertainty-Error Correlation	0.52
Coverage within $\mu \pm \sigma$	0.64

The performance of the volume prediction model is summarized in Table VI. The model achieves a mean absolute error of 0.640 Å³/atom, demonstrating that lattice volume can be estimated accurately using composition-only features combined with physics-informed descriptors.

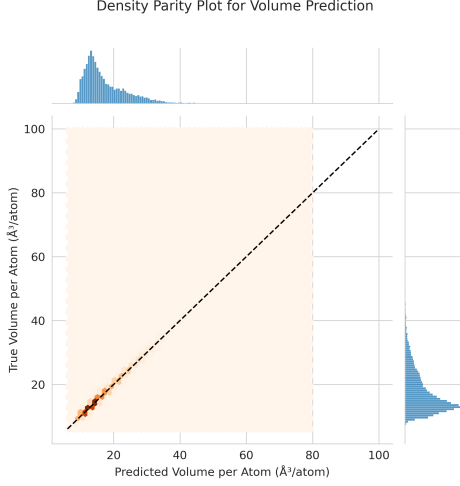


Fig. 6: Parity plot for volume per atom prediction.

The parity plot for volume prediction is shown in Figure 6, where most predictions lie close to the diagonal line, indicating accurate regression performance.

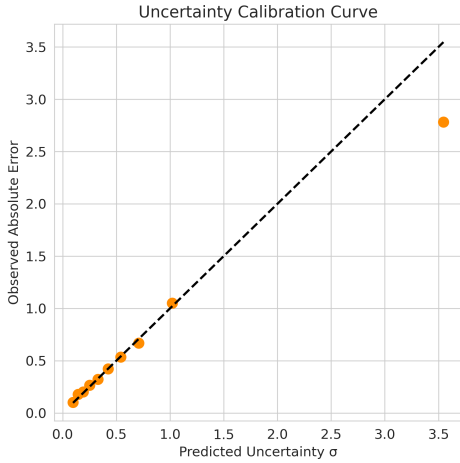


Fig. 7: Uncertainty versus prediction error.

The relationship between prediction uncertainty and absolute error is shown in Figure 7, demonstrating that uncertainty estimates correlate with prediction error.

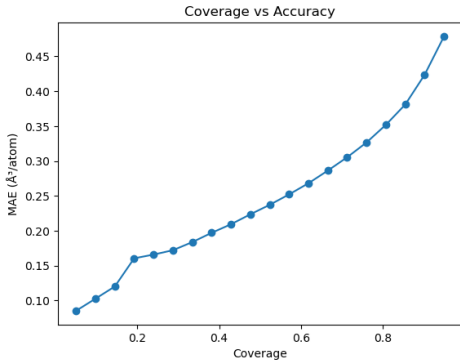


Fig. 8: Coverage versus accuracy curve.

The coverage versus accuracy curve shown in Figure 8 indicates that removing high-uncertainty predictions significantly reduces prediction error.

D. Space Group Prediction

Multiple machine learning models were evaluated for space group prediction, including Random Forest, XGBoost, and neural network models. The dataset contained 101,579 materials extracted from the Materials Project database.

TABLE VII: Space Group Prediction Model Comparison

Model	Top-1 Accuracy	Top-3 Accuracy	Top-5 Accuracy
Random Forest	0.662	0.836	0.882
Random Forest Ensemble	0.655	0.835	0.881
XGBoost	0.475	0.767	0.855
MLP	0.386	0.677	0.782
RF + Crystal System Filtering	0.772	0.894	0.919

The comparison of space group prediction models is shown in Table VII. As shown in Table VII, Random Forest models outperform neural networks and gradient boosting models for this task. The highest accuracy is achieved when space group predictions are constrained using crystal system filtering.

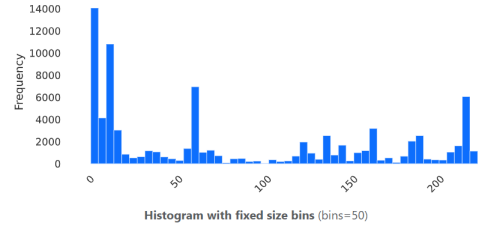


Fig. 9: Distribution of space groups in the dataset.

The distribution of space groups in the dataset is shown in Figure 9.

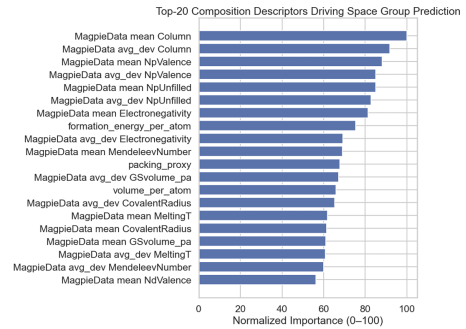


Fig. 10: Feature importance for space group prediction.

The feature importance plot for space group prediction is shown in Figure 10.

Instead of plotting the full 230-class matrix, Figure 11. truncates the predictions to the top 50 most frequent space groups. The dark-themed, row-normalized

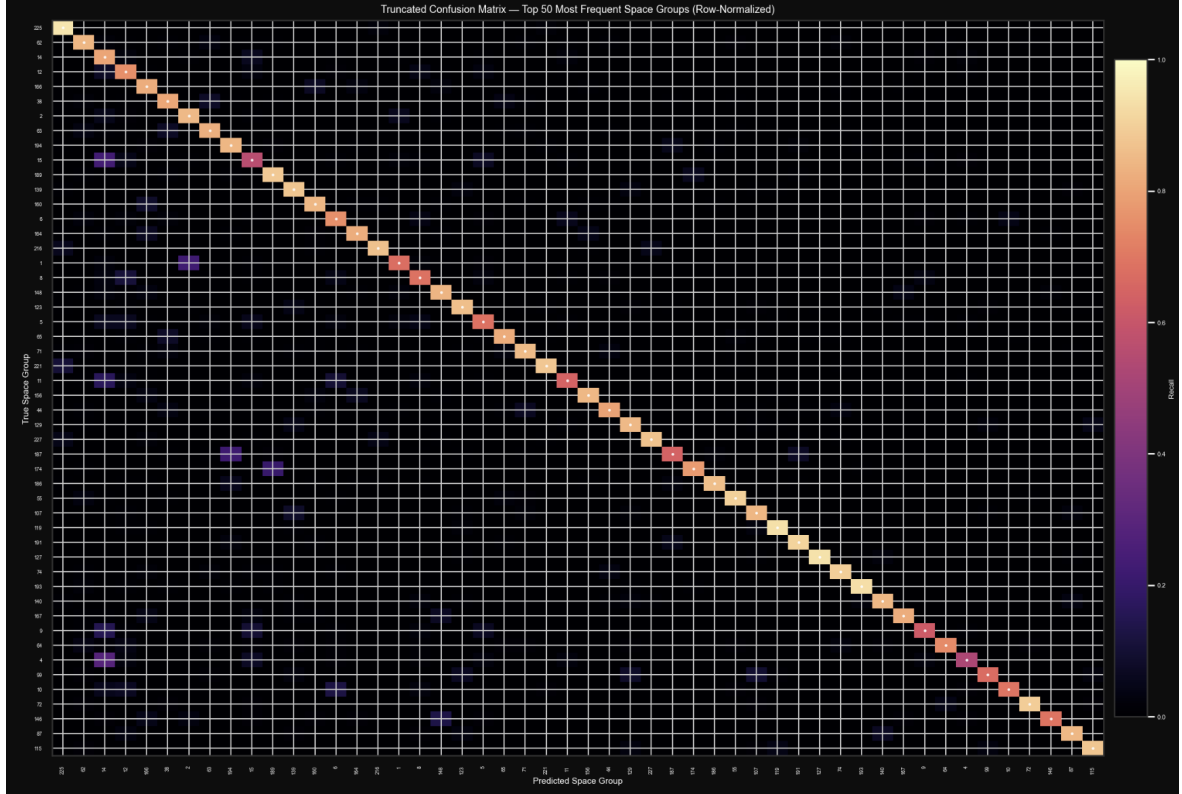


Fig. 11: Truncated Confusion Matrix — Top 50 Most Frequent Space Groups (Row-Normalized)

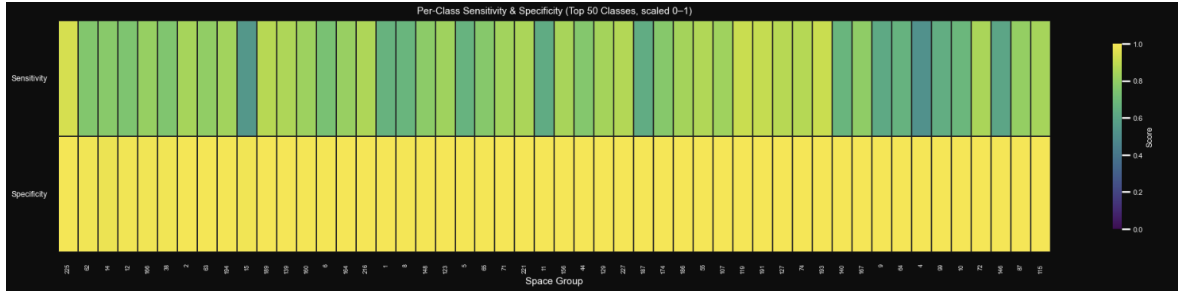


Fig. 12: Per-Class Sensitivity and Specificity (Top 50 Classes, scaled 0—1)

confusion matrix reveals that the Random Forest ensemble maintains a strong predictive diagonal for majority classes (e.g., Space Groups 225, 62, 14). The off-diagonal illuminations quickly diagnose where the model systematically confuses structurally similar orthorhombic or monoclinic symmetries due to overlapping compositional features.

To further dissect the model’s handling of the imbalanced dataset, Figure 12. presents a 1D heatmap of Per-Class Sensitivity and Specificity, directly mirroring the diagnostic approach used in recent structure-agnostic literature. The plot demonstrates that the model achieves near-perfect Specificity (true negative rate) across all top 50 classes. However, the Sensitivity (true positive rate) shows visible variance, confirming that the model’s ability to correctly recall a specific space group degrades as the number of training samples for that specific class decreases.

E. Wyckoff Position Prediction

Wyckoff prediction was performed using multi-label classification models, including LightGBM and XGBoost models. The dataset contained 210,579 crystal structures extracted from the Materials Project database.

TABLE VIII: Wyckoff Model Comparison

Model	Micro-F1	Macro-F1	Recall@3	Recall@5
LightGBM	0.84	0.71	0.79	0.91
XGBoost	0.85	0.73	0.81	0.92

The performance of Wyckoff prediction models is summarized in Table VIII. As shown in Table VIII, the XGBoost model achieves the best performance across all evaluation metrics.

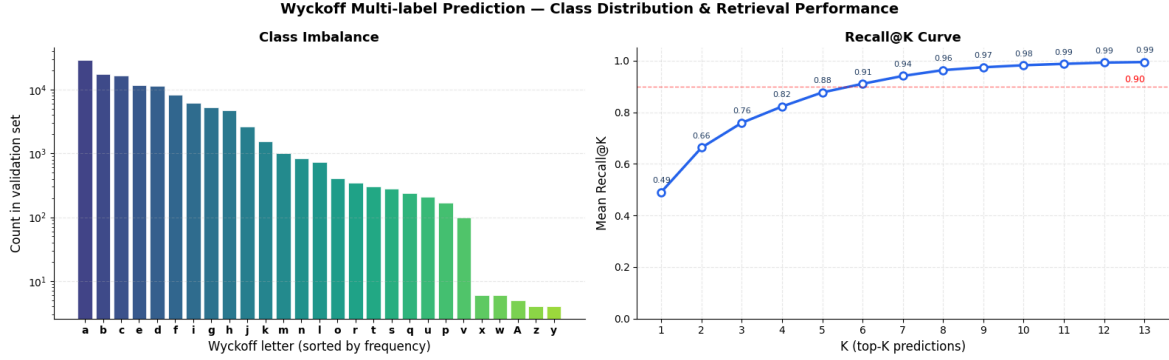


Fig. 13: Class Imbalance and Recall@K Curve.

The distribution of Wyckoff positions in the dataset is shown in Figure 13, which highlights the severe class imbalance in crystallographic data (for example, the letter ‘a’ has more than 10^4 instances, while ‘z’ and ‘y’ each appear only a few times). Furthermore, the Recall@K curve starts at 0.49 for $K = 1$ and increases rapidly, reaching 0.91 by $K = 6$ and approximately 0.99 by $K = 13$.

F. Overall Framework Summary

TABLE IX: Final Summary of All Models

Property	Best Model	Metric	Result
Formation Energy	Uncertainty Ensemble	MAE	0.0975 eV/atom
Hull Distance	Feature-Based MTL	MAE	0.059 eV
Stability	Uncertainty Ensemble	F1	0.9306
Volume	UQ Volume Model	MAE	0.640 Å ³ /atom
Space Group	RF + Crystal System	Top-1	0.772
Wyckoff	XGBoost	Micro-F1	0.85

The overall performance of the proposed framework is summarized in Table IX. The results demonstrate that composition-only machine learning models can predict thermodynamic and structural descriptors with good accuracy.

To rigorously evaluate the physical realism of our predicted Wyckoff configurations, we adopted the discrete probability validation method utilized in the ShotgunCSP framework. Rather than relying solely on classification accuracy, this method evaluates whether the generative model’s output probability distribution matches the actual relative frequencies of Wyckoff letters found in stable materials.

As shown in the top panel of Figure 14, the mean predicted soft probabilities from our structure-aware ensemble align remarkably well with the ground truth frequencies for the dominant Wyckoff letters (such as ‘a’, ‘b’, ‘c’, and ‘d’). This indicates that the model has successfully learned to map complex stoichiometric ratios to valid crystallographic symmetries rather than outputting random noise.

G. Discussion

Results indicate that the chemical compositions of crystalline materials can directly predict a number of thermodynamic and structural properties, using machine learning models. The use of multi-task learning framework to predict thermodynamic properties has increased prediction accuracy by learning the common representation for formation energy, energy above hull and stability, three physically related properties of crystalline material. The predictive performance of ensemble learning methods has improved by increasing prediction robustness and providing uncertainty information. The correlation of prediction error and uncertainty allows for risk-controlled screening of new materials [12].

By including physics-based descriptors, such as atomic radius, electronegativity, and the number of valence electrons, the accuracy of predicted crystal volume was substantially improved; demonstrating the necessity of including meaningful/physical descriptors in the model. Tree-based models achieved higher prediction accuracy compared to neural networks when predicting space group’s symmetry due to the ability to learn the non-linear relationships within the composition features. Inclusion of space group’s symmetry conditions when predicting Wyckoff’s position led to accurate predictions of their positions for the physically plausible crystal structures.

The authors provide a hierarchical method of predicting volume, symmetry and Wyckoff position of crystalline materials sequentially from their chemical compositions. The development of this method paves the way for the characterizing crystal structures of materials based on their chemical composition, without having to generate crystal structure from DFT calculations and providing researchers opportunities conduct large-scale material discovery.

VI. CONCLUSION

Our work illustrates the ability to predict thermodynamic and structural characteristics of inorganic crystalline materials through a composition-based machine learning (ML) architecture. The predicted prop-

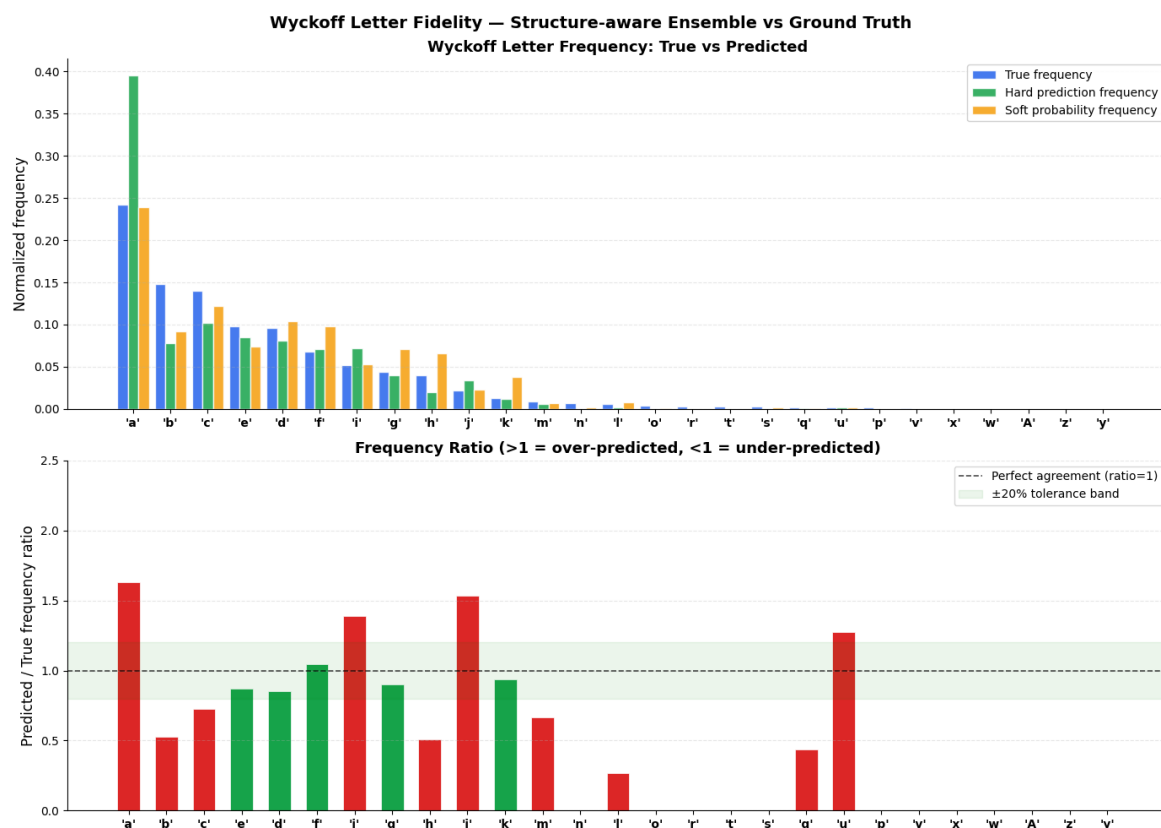


Fig. 14: Wyckoff Letter Fidelity

erties include formation energy, energy above hull, relative stability, volume per atom, space group, and Wyckoff positions. The predictions generated utilizing a composition-based feature set, combined with multi-task learning, ensemble techniques, and physics-informed descriptors yielded accurate property predictions. This hierarchy of prediction framework allows for the direct physical prediction of structural descriptors using composition without having to generate crystal structures or perform density functional theory (DFT) calculations. The proposed method will allow for large-scale screening of materials and form the foundation for future studies in crystal structure prediction and inverse material design.

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